

Semi-Direct Product

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Suppose we have two subgroups N and H of a group G . We have a canonical map $f : N \times H \rightarrow G$ (sending (n, h) to their product nh in G).

Problem 1. *This is a homomorphism iff N and H commute.*

Proof. $\implies$: Suppose f is a homomorphism. Then by definition, $\forall n \in N, h \in H, f((n, h)) = f((e, h) \cdot (n, e)) = nh = hn = f((e, h)) \cdot f((n, e))$. Thus $HN = NH$.

$\impliedby$: Suppose $HN = NH$, then $\forall n_1, n_2 \in N, h_1, h_2 \in H, f((n_1, h_1) \cdot (n_2, h_2)) = f((n_1 n_2, h_1 h_2)) = n_1 n_2 h_1 h_2 = n_1 h_1 n_2 h_2 = f((n_1, h_1)) f((n_2, h_2))$. Thus f is a homomorphism. $\square$

Problem 2. *It is injective iff $N \cap H = e$*

Proof. $\implies$: f is injective. $\forall x \in N \cap H, f(x, e) = x, f(e, x) = x$. Since f is injective, $(x, e) = (e, x)$ thus $x = e$. Thus $N \cap H = \{e\}$

$\impliedby$: $N \cap H = \{e\}$. $\forall (n_1, h_1), (n_2, h_2) \in N \times H \ni f(n_1, h_1) = f(n_2, h_2)$. Thus $n_1 h_1 = n_2 h_2 \implies n_2^{-1} n_1 = h_2 h_1^{-1}$. $n_2^{-1} n_1 \in N$ and $h_2 h_1^{-1} \in H$. Thus $n_2^{-1} n_1 = h_2 h_1^{-1} = e \in N \cap H$. Thus $n_1 = n_2$ and $h_2 = h_1$. Thus f is injective. $\square$

Problem 3. *It is surjective iff $NH = G$*

Proof. $\implies$: f is surjective. $\forall g \in G, \exists (n, h) \in N \times H \ni f(n, h) = g$ but $f(n, h) = nh = g$. Thus $\forall g \in G, \exists n \in N, h \in H \ni nh = g$. Thus $NH = G$

$\impliedby$: $NH = G$. Thus $\forall g \in G, \exists n \in N, h \in H \ni nh = g$. $f(n, h) = nh = g$. Thus f is surjective. $\square$

Problem 4. *Take S_3 , and $A_3 = N$ and $H = \langle (12) \rangle$. Check we get a bijection, but not an iso —(if it was an iso, S_3 would be Abelian). So what is the mult rule that S_3 puts on $N \times H$?*

Proof. $N \cap H = \{()\}$ as all other elements in H are two cycles which are not even permutations and thus are not in A_3 . Thus $f : N \times H \rightarrow G$ is injective. $\forall \phi \in S_3, \phi = x^a y^b$ for $x \in A_3$ and $y \in H, a \in [0, 1, 2]$ and $b = [0, 1]$. Thus $\phi = f(x^a, y^b)$. Thus f is surjective and a bijection exists from $N \times H \rightarrow G$. However in $S_3 \cong D_3, nh \neq hn$ but rather $hn = n^{-1}h$. Thus $NH \neq HN$ and f is not a homomorphism. S_3 puts the multiplication rule that $hn = n^{-1}h$ on $N \times H$. $\square$

Problem 5. Show that G is completely determined by N , H and the restriction of this conjugation map, $H \rightarrow \text{Aut}(N)$:

Hint: we know from the bijection that every element of G has a unique expression as a product nh . What we have to figure out is how to write

$n_1h_1n_2h_2 = n_3h_3$ (we know that the left hand side has a unique expression as on the right, the question is what that expression is) Note

$$h_1n_2 = h_1n_2h_1^{-1}h_1 = C_{h_1}(n_2)h_1 \text{ and thus}$$

$n_1h_1n_2h_2 = n_1C_{h_1}(n_2)h_1h_2 = n_3h_3$. Put another way, the "new" multiplication rule on the set $N \times H$ is given by

$(n_1, h_1)(n_2, h_2) := (n_1C_{h_1}(n_2), h_1h_2)$ – so the h 's multiply the usual way, but in order to "move the h_1 past the n_2 " we have to change n_2 , by conjugating by h_1 (note this produces another element of N).

Proof. $f : N \times H \rightarrow G$ is a bijection. So $\forall g \in G$, $g = f(n_3, h_3) = f((n_1, h_1) \cdot (n_2, h_2)) = n_1h_1n_2h_2 = n_3h_3$. $h_1n_2 = h_1n_2h_1^{-1}h_1 = C_{h_1}(n_2)h_1$. Thus $n_1h_1n_2h_2 = n_1C_{h_1}(n_2)h_1h_2$, $n_3 = n_1C_{h_1}(n_2) \in N$, thus $g = n_3h_3$ where $h_3 = h_1h_2$.

Thus define $(n_1, h_1) \cdot (n_2, h_2) := (n_1C_{h_1}(n_2), h_1h_2)$

Thus every element in G can be written as some n and h and thus G is completely determined by N , H and restriction of $H \rightarrow \text{Aut}(N)$. $\square$

Problem 6a. Show that the product set, with this mult rule, is a group.

Proof. 1. **Closure:** $(n_1, h_1), (n_2, h_2) \in N \times H$, $(n_1, h_1) \cdot (n_2, h_2) = (n_1C_{h_1}(n_2), h_1h_2)$, $n_1C_{h_1}(n_2) \in N$ and $h_1h_2 \in H$. Thus $(n_1C_{h_1}(n_2), h_1h_2) \in N \times H$.

2. **Identity:** $(e, e) \in N \times H$, $(n, h) \in N \times H$, $(n, h) \cdot (e, e) = (n, h)$, $(e, e) \cdot (n, h) = (n, h)$

3. **Inverses:** $\forall (n, h) \in N \times H$, $(n, h) \cdot (C_{h^{-1}}(n^{-1}), h^{-1}) = (nC_h(C_{h^{-1}}(n^{-1})), e) = (e, e)$ and $(C_{h^{-1}}(n^{-1}), h^{-1}) \cdot (n, h) = (C_{h^{-1}}(n^{-1})C_{h^{-1}}(n), e) = (e, e)$.

4. **Associativity:** $\forall (n_1, h_1), (n_2, h_2), (n_3, h_3) \in N \times H$.

$$((n_1, h_1) \cdot (n_2, h_2)) \cdot (n_3, h_3) = (n_1C_{h_1}(n_2), h_1h_2) \cdot (n_3, h_3) = (n_1C_{h_1}(n_2)C_{h_1h_2}(n_3), h_1h_2h_3)$$

$$(n_1, h_1) \cdot ((n_2, h_2) \cdot (n_3, h_3)) = (n_1, h_1) \cdot (n_2C_{h_2}(n_3), h_2h_3) = (n_1C_{h_1}(n_2C_{h_2}(n_3)), h_1h_2h_3)$$

$$(n_1C_{h_1}(n_2C_{h_2}(n_3)), h_1h_2h_3) = (n_1C_{h_1}(n_2)C_{h_1h_2}(n_3), h_1h_2h_3)$$

$\square$

Problem 6b. Show that there are natural copies of N and H in this group, with N normal

Proof. $\forall n \in N$, $(n, e) \in N \times H$ and $N \cong (N, e)$. $\forall h \in N$, $(e, h) \in N \times H$ and $H \cong (e, H)$, $\forall (a, b) \in N$, $\forall n \in N$, $(a, b)(n, e)(C_{b^{-1}}(a^{-1}), b^{-1}) = (aC_b(n), b)(C_{b^{-1}}(a^{-1}), b^{-1}) = (aC_b(n)a^{-1}, e) \in N$ because $aC_b(n)a^{-1} \in N$. $\square$

Problem 6c. G is called the semi-direct product the notation is roughly $N \rtimes_C H$. There are lots of examples. E.g. we can take N to be a cyclic group of order n , with generator X , and H to be a cyclic group of order 2, with generator y , and define $C_y : N \rightarrow N$ by $C_y(X) = X^{-1}$. What group do we get?

Proof. We get the group D_n . $N \times H \rightarrow D_n$ is a bijection. $N \cap H = \{e\}$. $\forall a \in D_n$ $a = x^c y^d$ where $a \in [0, \dots, n-1]$ and $b = 0, 1$. $N \rtimes_C H = \langle x, y | x^n = y^2 = e, yxy^{-1} = x^{-1} \rangle \cong D_n$. $\square$

Problem 7. As another example, figure out the possibilities with $N = F_2^2 = Z/2Z + Z/2Z$ and $H = \mu_3$. Note that $\text{Aut}(N) = GL_2(F_2) = S_3$. So we need a homomorphism of H to $GL_2(F_2) = S_3$. If we write $H = \langle y \rangle$, Y a generator, then such a homomorphism is the same as an element A in GL_2 with $A^3 = e$. If A is trivial, we will get the ordinary product. Otherwise there are two choices, which are squares of each other – figure out why the two choices give isomorphic semi direct products. Write down the resulting group in terms of generators and relations.

Proof. $\text{Aut}(N) = GL_2(F_2) = S_3 = \text{Perm}(F_2^2 \setminus \{0\})$. $C : H \rightarrow GL_2(F_2)$. y must go to an element in $a \in S_3$ such that $a^3 = e$ so $a = 1, (123), (132)$. If $a = e$, conjugation by a would result in a normal

product. $X = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

$$Y = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$C_1(y) = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} y \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^{-1} \text{ or } C_2(y) = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} y \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{-1}.$$

$$a = e_1, b = e_2, c = e_1 + e_2.$$

$$C_1(a) = b, C_1(b) = ab$$

$$H \times_{C_1} N = \langle a, b, X : a^2 = b^2 = e, ab = ba, XaX^{-1} = b, XbX^{-1} = ab \rangle$$

$$C_2(c) = b, C_2(b) = bc$$

$$H \times_{C_1} N = \langle c, b, X : a^2 = b^2 = e, ab = ba, XcX^{-1} = b, XbX^{-1} = bc \rangle$$

They are isomorphic because it is only up to choice of generators or the basis of F_2^2 . $\square$

Problem 8. Not every group is a semi-direct product of simpler groups – e.g. a simple group (one with no normal subgroups) can't be. Check that the quaternion group Q_8 is not (in a non-trivial way) a semi-direct product (check that every non-trivial subgroup contains the center, so we can never satisfy 2 and 3). But "most" groups of small order are.

Proof. Let $H \leq Q_8$ be any subgroup where $H \neq \{1\}$. If $Z(Q_8) = H$, we are done.

If $-1 \in H$, then $\langle -1 \rangle = Z(Q_8) \subset H$.

If $x \in H \ni x \in \{\pm i, \pm j, \pm k\}$, by definition, $x^2 = -1$ so $-1 \in H$ thus the previous case is invoked.

So any nontrivial $H \leq Q_8$ has $Z(Q_8) \subset H$

So $\forall H, N < G$. If H, N are nontrivial, $Z(Q_8) \subset H \cap N$ thus $H \times N \rightarrow G$ is not injective. If one subgroup is trivial, then the other one is G . But since they are both strictly proper subgroups this forms a contradiction and the map is not surjective. Thus Q_8 cannot be a semidirect product as it is not in bijection with the product of any of its subgroups. $\square$

Problem 9a. Here is a quite cool question that might stretch your head: Fix H and N , Let $A := \text{Aut}(N)$ (group automorphisms, of course). Now take two C_1, C_2 in $\text{Hom}_{\text{groups}}(H, A)$. We can use them to form two different semi-direct products G_1 and G_2 , each containing canonical copies of H and N , so that the natural map $N_i \times H_i \rightarrow G_i$ is a bijection.

When is there an isomorphism $G_1 \rightarrow G_2$ taking N_1 to N_2 , H_1 to H_2 ? Hint: Note that A and $\text{Aut}(H)$ each naturally act on $\text{Hom}_{\text{groups}}(H, A)$: If we have a in A , and f in $\text{Aut}(H)$ and $g : H \rightarrow A$, we can take $f(g) := g \circ f^{-1}$, and $a(g) := C_a \circ g$, where $C_a : A \rightarrow A$ is conjugation by a .

Problem 9b. Q : Why the inverse on f ?